

# MediVault: AI powered reports scanner and personal Health locker

An intelligent, secure personal health information system combining an advanced mobile frontend with real-time Generative AI medical document parsing and automated vital telemetry tracking.

## 1. Brief Overview

MediVault is designed to solve the problem of fragmented and complex clinical documentation. It provides a personal medical locker that allows patients to snap photos of physical clinical documents and immediately parse them into structured, trackable health profiles in real time. It serves as a unified baseline for critical medical metrics, chronic diagnoses, and consulting references.

## 2. Core Features

- **Real-Time Multimodal Camera Scanner:** Captures high-fidelity photos of medical reports directly inside the app, uploading compressed base64 images securely to the backend.
- **Multimodal Visual OCR Engine:** Feeds photo assets directly to Google Gemini 2.0 Flash to perform deep document text extraction, completely bypassing the need for native offline OCR libraries.
- **AI Clinical Summaries:** Generates automated, patient-friendly, easy-to-read clinical explanations under a read-only lock to preserve data integrity.
- **Profile Smart-Sync Updates:** Parses critical laboratory findings (blood glucose, blood pressure, pulse, chronic conditions, and allergies) and automatically updates the patient's master clinical baseline.
- **Telemetry Metrics & Trends:** Automatically logs extracted vital metrics into SQLite database tables on the report date, generating dynamic 90-day tracking logs.

## 3. Technological Stack & Tools

| Layer           | Technologies & Tools                                                              |
|-----------------|-----------------------------------------------------------------------------------|
| Mobile Frontend | React Native, Expo SDK 54, Hermes Engine, Zustand, Lucide Icons, React Navigation |
| Backend Server  | Node.js, Express, TypeScript, ts-node-dev                                         |

| Layer          | Technologies & Tools                                              |
|----------------|-------------------------------------------------------------------|
| Database & ORM | SQLite Database, Prisma ORM, Prisma Client                        |
| AI Engine      | Google Generative AI SDK, Gemini 2.0 Flash, Gemini Multimodal API |

### Authentication Architecture & Sandbox Pattern

To support rapid, deterministic sandbox testing, a Single-User Sandbox Pattern was engineered. During registration or login, the database truncates all SQLite tables to provide a fresh canvas. For production environments, this will transition to dynamic UUIDv4 keys, bcrypt password salting (minimum 10 rounds), and secure signed JSON Web Tokens (JWT) passed in client Authorization headers.

## 4. Future Roadmap Ideas

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- **Edge-AI Local Scanning:** Integrate local on-device LLMs (such as Gemini Nano) to execute visual OCR directly on-device without requiring network availability.
- **Health Ecosystem Synchronization:** Build native bridges to automatically synchronize parsed vital telemetry with Apple HealthKit and Google Fit databases.
- **Paramedic Offline QR Access:** Create locked-screen emergency QR code widgets containing encrypted key baseline metrics that first responders can scan offline instantly.

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